

Gradient Descent Method

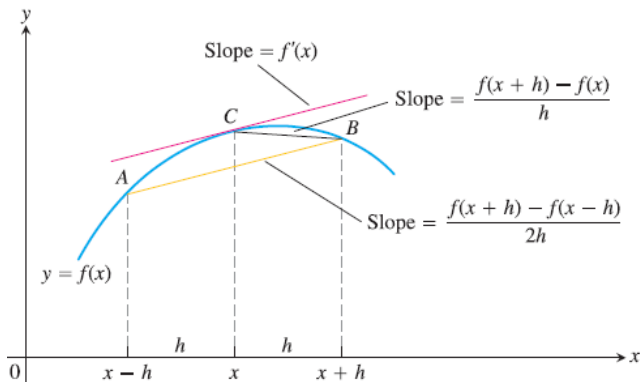
Dr. Yasir Ali (yali@ceme.nust.edu.pk)

DBS&H, CEME-NUST

November 26, 2025

Derivative of $f(x)$ at point a **along x-axis**

$$f'(x)|_{\text{at } x=a} = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$



Derivative of $f(x)$ at point a **along x-axis**

$$f'(x)|_{\text{at } x=a} = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

Derivative of $f(x)$ at point a **along x-axis**

$$f'(x)|_{\text{at } x=a} = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

Partial Derivative of $f(x, y)$ at point (a, b) **along x-axis**

$$\frac{\partial f}{\partial x}(a, b) = \lim_{h \rightarrow 0} \frac{f(a+h, b) - f(a, b)}{h}$$

Derivative of $f(x)$ at point a **along x-axis**

$$f'(x)|_{\text{at } x=a} = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

Partial Derivative of $f(x, y)$ at point (a, b) **along x-axis**

$$\frac{\partial f}{\partial x}(a, b) = \lim_{h \rightarrow 0} \frac{f(a+h, b) - f(a, b)}{h}$$

Partial Derivative of $f(x, y)$ at point (a, b) **along y-axis**

$$\frac{\partial f}{\partial y}(a, b) = \lim_{k \rightarrow 0} \frac{f(a, b+k) - f(a, b)}{k}$$

Derivative of $f(x)$ at point a **along x-axis**

$$f'(x)|_{\text{at } x=a} = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

Partial Derivative of $f(x, y)$ at point (a, b) **along x-axis**

$$\frac{\partial f}{\partial x}(a, b) = \lim_{h \rightarrow 0} \frac{f(a+h, b) - f(a, b)}{h}$$

Partial Derivative of $f(x, y)$ at point (a, b) **along y-axis**

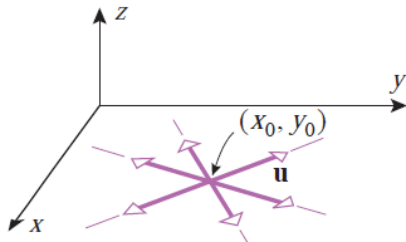
$$\frac{\partial f}{\partial y}(a, b) = \lim_{k \rightarrow 0} \frac{f(a, b+k) - f(a, b)}{k}$$

Partial Derivative of $f(x, y)$ at point (a, b) **along unit vector $u = [u_1, u_2]$**

$$D_u f(a, b) = \lim_{h \rightarrow 0} \frac{f(a + hu_1, b + hu_2) - f(a, b)}{h}$$

Infinitely Many Possibilities

To find how $f(x, y)$ changes from (x_0, y_0) in a given direction, we need a simple way to describe that direction among infinitely many possibilities.



One way to do this is to use a unit vector

$$\hat{u} = u_1 \hat{i} + u_2 \hat{j}$$

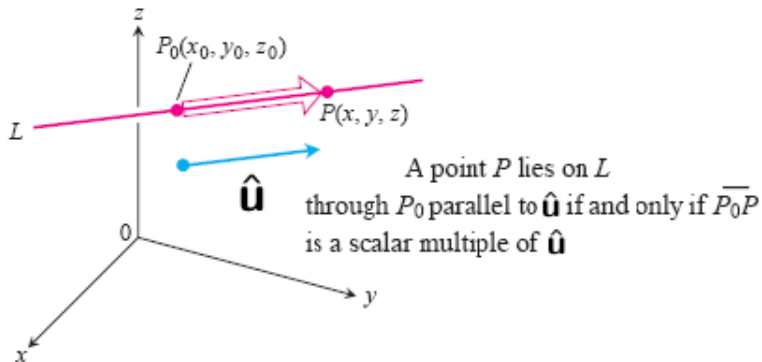
The initial point at (x_0, y_0) and points in the desired direction.

A unit vector $\hat{u} = [u_1, u_2]$ defines a line through (x_0, y_0) as

$$x = x_0 + tu_1, \quad y = y_0 + tu_2,$$

where t is arc length from (x_0, y_0) , increasing along u .

Parametric equations of straight line passing through point (x_0, y_0, z_0) and parallel to $\hat{u} = [u_1, u_2, u_3]$



$$\begin{aligned}x &= x_0 + u_1 t \\y &= y_0 + u_2 t \\z &= z_0 + u_3 t\end{aligned} \quad (t \in \mathbb{R}).$$

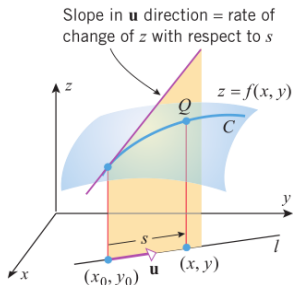
$$\underbrace{r(t) = r_0 + ut}_{\text{Vector Form}}$$

Along the line, $z = f(x_0 + tu_1, y_0 + tu_2)$ is a function of t . The derivative $\frac{dz}{dt}$ at $t = 0$ gives the rate of change of $f(x, y)$ from (x_0, y_0) in direction u .

Directional Derivative

The directional derivative of $f(x, y)$ at (x_0, y_0) in direction $\hat{u}$ is

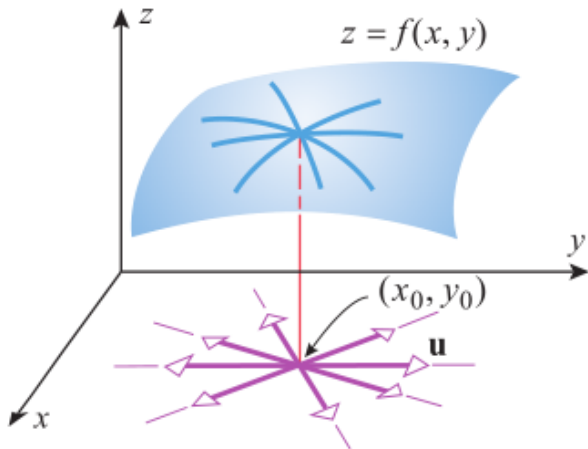
$$D_u f(x_0, y_0) = \left. \frac{d}{dt} f(x_0 + tu_1, y_0 + tu_2) \right|_{t=0}, \quad \text{if the derivative exists.}$$



Geometrically

$D_u f(x_0, y_0)$ is the slope of the surface $z = f(x, y)$ at $(x_0, y_0, f(x_0, y_0))$ in direction u .

It depends on both the point and the direction.



The slope of the surface varies with the direction of $\mathbf{u}$.

Moving in any direction u

$$x = x_0 + tu$$

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a continuous function. Consider $x \in \mathbb{R}^n$ and $a \in \mathbb{R}^n$. The directional derivative of f in x in the direction u is given by

$$\lim_{t \rightarrow 0} \frac{f(x + tu) - f(x)}{t}$$

The directional derivative of $f(x, y)$ at (x_0, y_0) in direction $\hat{u}$ is

$$\begin{aligned} D_u f(x_0, y_0) &= \left. \frac{d}{dt} f(x_0 + tu_1, y_0 + tu_2) \right|_{t=0}, \quad \text{if the derivative exists.} \\ &= \frac{\partial f}{\partial x}(x_0, y_0)u_1 + \frac{\partial f}{\partial y}(x_0, y_0)u_2 = f_x(x_0, y_0)u_1 + f_y(x_0, y_0)u_2 \\ &= (f_x \hat{i} + f_y \hat{j}) \cdot (u_1 \hat{i} + u_2 \hat{j}) = \nabla f \cdot u \end{aligned}$$

Directional Derivatives

$$D_u f(x, y) = \nabla f \cdot u$$

The gradient of $f(x, y)$ is the vector-valued function

$$\nabla f = [f_x, f_y] = f_x \hat{i} + f_y \hat{j}$$

$$\nabla f = [f_x, f_y, f_z] = f_x \hat{i} + f_y \hat{j} + f_z \hat{k}$$

Directional Derivative of a Scalar $f(x, y)$ in u Direction

$$D_u f = \nabla f \cdot u = f_x u_1 + f_y u_2$$

$$D_u f = \nabla f \cdot u = f_x u_1 + f_y u_2 + f_z u_3$$

$$D_u f = \nabla f \cdot \hat{u} = |\nabla f| |\hat{u}| \cos \theta \Rightarrow \boxed{D_u f = |\nabla f| \cos \theta} \because |\hat{u}| = 1$$

For $f(x, y) = x^2y - 4y^3$, compute $D_u f(2, 1)$ for the directions

- 1 $u = [\frac{\sqrt{3}}{2}, \frac{1}{2}]$
- 2 u in the direction from $(2, 1)$ to $(4, 0)$.

$$D_u f = f_x u_1 + f_y u_2, \text{ where } u = [\overset{u_1}{\frac{\sqrt{3}}{2}}, \overset{u_2}{\frac{1}{2}}]$$

- 1 $f_x = 2xy$ and $f_y = x^2 - 12y^2$. $f_x(2, 1) = 4$, $f_y(2, 1) = -8$

$$D_u f(2, 1) = 4\left(\frac{\sqrt{3}}{2}\right) - 8\left(\frac{1}{2}\right) = 2\sqrt{3} - 4$$

- 2 unit vector from $(2, 1)$ to $(4, 0)$ is $v = [\frac{2}{\sqrt{5}}, \frac{-1}{\sqrt{5}}]$.

$$D_v f(2, 1) = 4\left(\frac{2}{\sqrt{5}}\right) - 8\left(\frac{-1}{\sqrt{5}}\right) = \frac{16}{\sqrt{5}}$$

For $f(x, y, z) = x^2y - yz^3 + z$, at the point $(1, -2, 0)$ in the direction of $\mathbf{v} = 2\hat{i} + \hat{j} - 2\hat{k}$

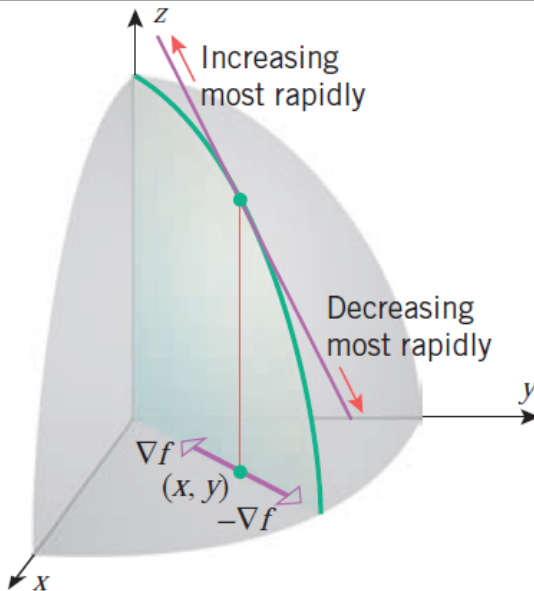
For $f(x, y, z) = x^2y - yz^3 + z$, at the point $(1, -2, 0)$ in the direction of $\mathbf{v} = 2\hat{i} + \hat{j} - 2\hat{k}$

$$D_{\mathbf{u}}f = f_x u_1 + f_y u_2 + f_z u_3, \text{ where } \mathbf{u} = \frac{\mathbf{v}}{|\mathbf{v}|} = \left[\frac{2}{3}, \frac{1}{3}, -\frac{2}{3}\right]$$

$$f_x = 2xy, \quad f_y = x^2 - z^3 \text{ and } f_z = -3yz^2 + 1.$$

$$f_x(1, -2, 0) = -4, \quad f_y(1, -2, 0) = 1, \quad f_z(1, -2, 0) = 1$$

$$D_{\mathbf{u}}f(1, -2, 0) = -4\left(\frac{2}{3}\right) + \frac{1}{3} - \frac{2}{3} = -3$$



- ① The function f increases most rapidly when $\cos \theta = 1$ or when $\theta = 0$ and $\mathbf{u}$ is the direction of ∇f . That is, at each point P in its domain, f increases most rapidly in the direction of the gradient vector ∇f at P . The derivative in this direction is

$$D_u f = \nabla f \cdot \mathbf{u} = |\nabla f| \cos 0 = |\nabla f|$$

- ② Similarly, f decreases most rapidly in the direction of $-\nabla f$. The derivative in this direction is

$$D_u f = |\nabla f| \cos \pi = -|\nabla f|$$

- ③ Any direction $\mathbf{u}$ orthogonal to a gradient $\nabla f \neq 0$ is a direction of zero change in f because $\theta = \frac{\pi}{2}$ and

$$D_u f = |\nabla f| \cos \frac{\pi}{2} = 0$$

Properties of $D_u f = \nabla f \cdot \mathbf{u} = |\nabla f| \cos \theta$

Property	Direction of $\hat{\mathbf{u}}$	Angle θ	$\cos \theta$	Rate of Change
Max. Increase	Same as ∇f	0°	1	$ \nabla f $
Zero Change	Perpendicular to ∇f	90°	0	0
Max. Decrease	Opposite to ∇f	180°	-1	$- \nabla f $

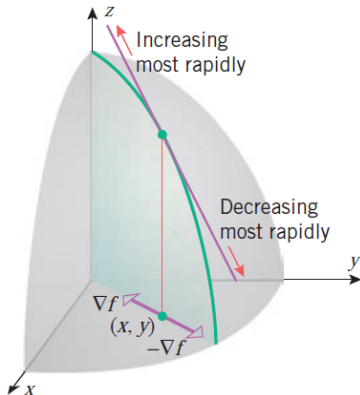
Key Insight

- ∇f points in direction of **steepest ascent**
- $-\nabla f$ points in direction of **steepest descent**
- Directions perpendicular to ∇f follow **level curves**

Quick Reference

If you want to...	Move in direction...	Rate of change
Increase fastest	∇f	$ \nabla f $
Decrease fastest	$-\nabla f$	$- \nabla f $
Stay at same level	$\perp \nabla f$	0

Visualizing Directional Derivatives



Angle θ between ∇f and u

Angle Dependence

$$D_u f = |\nabla f| \cos \theta$$

- $0^\circ < \theta < 90^\circ$: Positive increase
- $\theta = 90^\circ$: Zero change
- $90^\circ < \theta < 180^\circ$: Negative decrease

Gradient Property

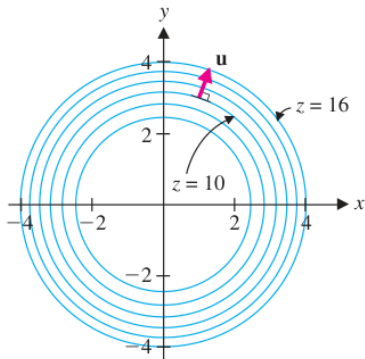
At each point, the gradient ∇f points in the direction of maximum increase of the function.

Find the maximum and minimum rates of change of the function $f(x, y) = x^2 + y^2$ at the point $(1, 3)$

$$\nabla f(x, y) = [2x, 2y], \quad \Rightarrow \nabla f(1, 3) = [2, 6]$$

Maximum Rate of Change: $|\nabla f|$ that is $\sqrt{2^2 + 6^2} = \sqrt{40}$

Max Change Occurs in Direction: $\hat{u} = \frac{\nabla f}{|\nabla f|}$ that is $\hat{u} = \left[\frac{2}{\sqrt{40}}, \frac{6}{\sqrt{40}} \right]$



Find the maximum and minimum rates of change of the function $f(x, y) = x^2 + y^2$ at the point $(1, 3)$

$$\nabla f(x, y) = [2x, 2y], \quad \Rightarrow \nabla f(1, 3) = [2, 6]$$

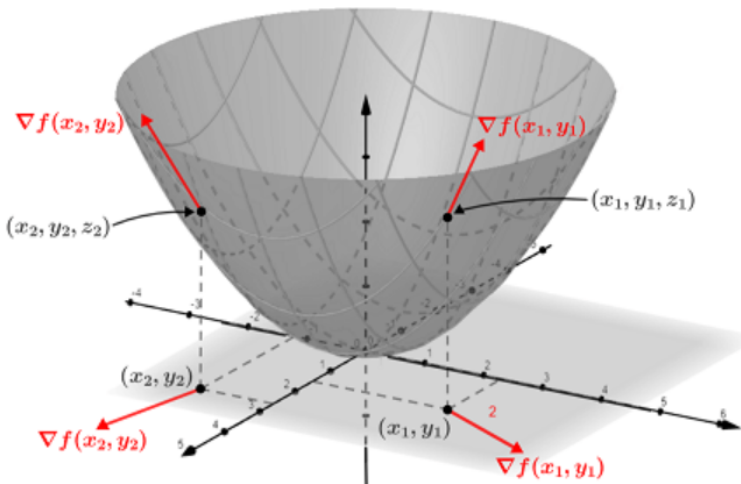
Rate of Change

- Maximum Rate of Change:
 $|\nabla f| = \sqrt{40}$
- Minimum Rate of Change:
 $-|\nabla f| = -\sqrt{40}$

Direction of Extreme Change

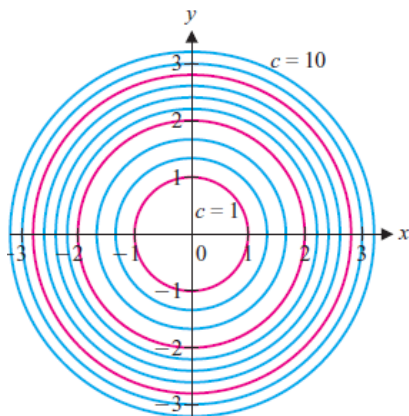
- Direction of Maximum Change:
 $\hat{u} = \frac{\nabla f}{|\nabla f|} = \left[\frac{2}{\sqrt{40}}, \frac{6}{\sqrt{40}} \right]$
- Direction of Minimum Change:
 $\hat{u} = -\frac{\nabla f}{|\nabla f|} = -\left[\frac{2}{\sqrt{40}}, \frac{6}{\sqrt{40}} \right]$

Example of Directional Derivative

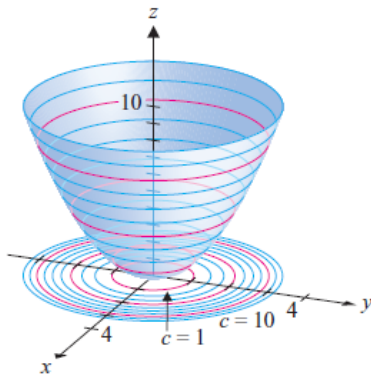


Level Curves or Contours

Level curves are traces in xy -planes $z = c$ of the surface $z = f(x, y)$.

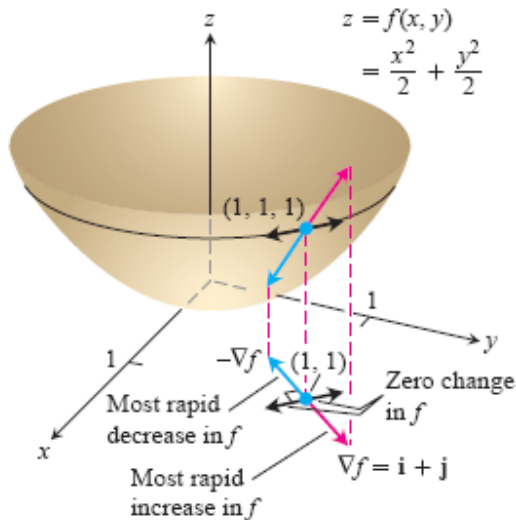


Contour plot of $g(x, y) = x^2 + y^2$



$z = x^2 + y^2$

Example of DD



Let $f(x, y) = x^2 e^y$. Find the maximum value of a directional derivative at $(-2, 0)$, and find the unit vector in the direction in which the maximum value occurs.

If $\nabla f \neq 0$ at P , then among all possible directional derivatives of f at P , the derivative in the direction of ∇f at P has the largest value. This value is $|\nabla f|$ at P .

$$\nabla f(x, y) = [f_x(x, y), f_y(x, y)] \implies \nabla f(x, y) = [2xe^y, x^2 e^y]$$

The Gradient of f at $(-2, 0)$ is

$$\nabla f(-2, 0) = [-4, 4], \quad \text{and} \quad |\nabla f| = 4\sqrt{2}$$

This maximum occurs in the direction of $\nabla f(-2, 0)$. The unit vector in this direction is

$$\mathbf{u} = \frac{\nabla f}{|\nabla f|} = \left[-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right]$$

Gradient Descent Method: Mathematical Foundation

Gradient Descent Method: Mathematical Foundation

Basic Gradient Descent Update Rule

$$\mathbf{x}_{k+1} = \mathbf{x}_k - \alpha \nabla f(\mathbf{x}_k)$$

Where:

- $\mathbf{x}_k$: Current parameters
- $\mathbf{x}_{k+1}$: Updated parameters
- α : Learning rate
- $\nabla f(\mathbf{x}_k)$: Gradient

Component-wise Form

$$x_1^{(k+1)} = x_1^{(k)} - \alpha \frac{\partial f}{\partial x_1}$$

$$x_2^{(k+1)} = x_2^{(k)} - \alpha \frac{\partial f}{\partial x_2}$$

$$\vdots$$

$$x_n^{(k+1)} = x_n^{(k)} - \alpha \frac{\partial f}{\partial x_n}$$

Gradient Descent Variants

Fixed Learning Rate Method

$$\mathbf{x}_{k+1} = \mathbf{x}_k - \alpha \nabla f(\mathbf{x}_k)$$

- α is constant throughout iterations
- Simple but requires careful tuning

Optimal Step Size (Line Search) Method

$$\mathbf{x}_{k+1} = \mathbf{x}_k - \lambda_k^* \nabla f(\mathbf{x}_k)$$

where λ_k^* is found by solving:
 $\lambda_k^* = \arg \min_{\lambda > 0} f(\mathbf{x}_k - \lambda \nabla f(\mathbf{x}_k))$

- Adaptive step size each iteration
- More computationally expensive but faster convergence

Search Direction Formulation

General Framework

$$\mathbf{s}_k = -\nabla f(\mathbf{x}_k) \quad (\text{Search direction})$$

$$\mathbf{x}_{k+1} = \mathbf{x}_k + \lambda_k \mathbf{s}_k$$

Fixed Step Size

$$\lambda_k = \alpha \quad (\text{constant})$$

- Low computational cost
- Many iterations needed

Optimal Step Size

$$\lambda_k = \lambda_k^* \quad (\text{line search})$$

- High computational cost
- Fewer iterations needed

Fixed Learning Rate Gradient Descent:

Step 1: Initialization

Start with an arbitrary initial point X_1 . Set the iteration number as $i = 1$.

Step 2: Search Direction Calculation

Find the search direction S_i as: $S_i = -\nabla f_i = -\nabla f(X_i)$

Step 3: Parameter Update

Update the parameters using the fixed learning rate α :

$$X_{i+1} = X_i + \alpha S_i = X_i - \alpha \nabla f_i$$

Step 4: Optimality Test

Test the new point, X_{i+1} , for optimality. If X_{i+1} is optimum, stop the process. Otherwise, go to step 5.

Step 5: Iteration Update

Set the new iteration number $i = i + 1$ and go to step 2.

Optimization Problem

Minimize: $f(x_1, x_2) = x_1 - x_2 + 2x_1^2 + 2x_1x_2 + x_2^2$

Starting point: $X_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

Gradient: $\nabla f = \begin{bmatrix} 1 + 4x_1 + 2x_2 \\ -1 + 2x_1 + 2x_2 \end{bmatrix}$

Method 1: Fixed Learning Rate ($\alpha = 0.1$)

Minimize: $f(x_1, x_2) = x_1 - x_2 + 2x_1^2 + 2x_1x_2 + x_2^2$

Gradient: $\nabla f(x_1, x_2) = \begin{bmatrix} 1 + 4x_1 + 2x_2 \\ -1 + 2x_1 + 2x_2 \end{bmatrix} \Rightarrow \nabla f_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

Iteration 1

- Current point: $X_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$
- Gradient: $\nabla f_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$
- Update: $X_2 = X_1 - \alpha \nabla f_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} - (0.1) \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} -0.1 \\ 0.1 \end{bmatrix}$
- Function value: $f(X_2) = -0.19$

Method 1: Fixed Learning Rate ($\alpha = 0.1$)

Minimize: $f(x_1, x_2) = x_1 - x_2 + 2x_1^2 + 2x_1x_2 + x_2^2$

Gradient: $\nabla f(x_1, x_2) = \begin{bmatrix} 1 + 4x_1 + 2x_2 \\ -1 + 2x_1 + 2x_2 \end{bmatrix} \Rightarrow \nabla f_2 = \begin{bmatrix} 0.6 \\ -0.8 \end{bmatrix}$

Iteration 2

- Current point: $X_2 = \begin{bmatrix} -0.1 \\ 0.1 \end{bmatrix}$
- Gradient: $\nabla f_2 = \begin{bmatrix} 0.6 \\ -0.8 \end{bmatrix}$
- Update: $X_3 = X_2 - \alpha \nabla f_2 = \begin{bmatrix} -0.1 \\ 0.1 \end{bmatrix} - (0.1) \begin{bmatrix} 0.6 \\ -0.8 \end{bmatrix} = \begin{bmatrix} -0.16 \\ 0.18 \end{bmatrix}$
- Function value: $f(X_3) = -0.3076$

Method 1: Fixed Learning Rate ($\alpha = 0.1$)

Minimize: $f(x_1, x_2) = x_1 - x_2 + 2x_1^2 + 2x_1x_2 + x_2^2$

Gradient: $\nabla f(x_1, x_2) = \begin{bmatrix} 1 + 4x_1 + 2x_2 \\ -1 + 2x_1 + 2x_2 \end{bmatrix} \Rightarrow \nabla f_3 = \begin{bmatrix} 0.36 \\ -0.6 \end{bmatrix}$

Iteration 3

- Current point: $X_3 = \begin{bmatrix} -0.16 \\ 0.18 \end{bmatrix}$
- Gradient: $\nabla f_3 = \begin{bmatrix} 0.36 \\ -0.6 \end{bmatrix}$
- Update: $X_4 = X_3 - \alpha \nabla f_3 = \begin{bmatrix} -0.16 \\ 0.18 \end{bmatrix} - (0.1) \begin{bmatrix} 0.36 \\ -0.6 \end{bmatrix} = \begin{bmatrix} -0.196 \\ 0.24 \end{bmatrix}$
- Function value: $f(X_4) = -0.3824$

Method 1: Fixed Learning Rate ($\alpha = 0.1$)

Minimize: $f(x_1, x_2) = x_1 - x_2 + 2x_1^2 + 2x_1x_2 + x_2^2$

Gradient: $\nabla f(x_1, x_2) = \begin{bmatrix} 1 + 4x_1 + 2x_2 \\ -1 + 2x_1 + 2x_2 \end{bmatrix} \Rightarrow \nabla f_4 = \begin{bmatrix} 0.216 \\ -0.472 \end{bmatrix}$

Iteration 4

- Current point: $X_4 = \begin{bmatrix} -0.196 \\ 0.24 \end{bmatrix}$
- Gradient: $\nabla f_4 = \begin{bmatrix} 0.216 \\ -0.472 \end{bmatrix}$
- Update: $X_5 = X_4 - \alpha \nabla f_4 = \begin{bmatrix} -0.196 \\ 0.24 \end{bmatrix} - (0.1) \begin{bmatrix} 0.216 \\ -0.472 \end{bmatrix} = \begin{bmatrix} -0.2176 \\ 0.2872 \end{bmatrix}$
- Function value: $f(X_5) = -0.4275$

$$\text{Minimize: } f(x_1, x_2) = x_1 - x_2 + 2x_1^2 + 2x_1x_2 + x_2^2$$

$$\text{Gradient: } \nabla f(x_1, x_2) = \begin{bmatrix} 1 + 4x_1 + 2x_2 \\ -1 + 2x_1 + 2x_2 \end{bmatrix}$$

i	X_i	∇f_i	Update X_{i+1}				$f(X_{i+1})$
1	$\begin{bmatrix} 0 \\ 0 \end{bmatrix}$	$\begin{bmatrix} 1 \\ -1 \end{bmatrix}$	$\begin{bmatrix} 0 \\ 0 \end{bmatrix}$	-0.1	$\begin{bmatrix} 1 \\ -1 \end{bmatrix}$	$= \begin{bmatrix} -0.1 \\ 0.1 \end{bmatrix}$	-0.19
2	$\begin{bmatrix} -0.1 \\ 0.1 \end{bmatrix}$	$\begin{bmatrix} 0.6 \\ -0.8 \end{bmatrix}$	$\begin{bmatrix} -0.1 \\ 0.1 \end{bmatrix}$	-0.1	$\begin{bmatrix} 0.6 \\ -0.8 \end{bmatrix}$	$= \begin{bmatrix} -0.16 \\ 0.18 \end{bmatrix}$	-0.3076
3	$\begin{bmatrix} -0.16 \\ 0.18 \end{bmatrix}$	$\begin{bmatrix} 0.36 \\ -0.6 \end{bmatrix}$	$\begin{bmatrix} -0.16 \\ 0.18 \end{bmatrix}$	-0.1	$\begin{bmatrix} 0.36 \\ -0.6 \end{bmatrix}$	$= \begin{bmatrix} -0.196 \\ 0.24 \end{bmatrix}$	-0.3824
4	$\begin{bmatrix} -0.196 \\ 0.24 \end{bmatrix}$	$\begin{bmatrix} 0.216 \\ -0.472 \end{bmatrix}$	$\begin{bmatrix} -0.196 \\ 0.24 \end{bmatrix}$	-0.1	$\begin{bmatrix} 0.216 \\ -0.472 \end{bmatrix}$	$= \begin{bmatrix} -0.2176 \\ 0.2872 \end{bmatrix}$	-0.4275
5	$\begin{bmatrix} -0.2176 \\ 0.2872 \end{bmatrix}$	$\begin{bmatrix} 0.1296 \\ -0.3792 \end{bmatrix}$	$\begin{bmatrix} -0.2176 \\ 0.2872 \end{bmatrix}$	-0.1	$\begin{bmatrix} 0.1296 \\ -0.3792 \end{bmatrix}$	$= \begin{bmatrix} -0.2306 \\ 0.3251 \end{bmatrix}$	-0.4563

Observations

- Steady, gradual progress toward minimum
- Consistent step size of $\alpha = 0.1$ throughout
- Function value decreases monotonically:
 $0 \rightarrow -0.19 \rightarrow -0.3076 \rightarrow -0.3824 \rightarrow -0.4275 \rightarrow -0.4563$

Method 2: Optimal Step Size (Line Search)

Minimize: $f(x_1, x_2) = x_1 - x_2 + 2x_1^2 + 2x_1x_2 + x_2^2$

Gradient: $\nabla f(x_1, x_2) = \begin{bmatrix} 1 + 4x_1 + 2x_2 \\ -1 + 2x_1 + 2x_2 \end{bmatrix} \Rightarrow \nabla f_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

Iteration 1

- Current point: $X_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$
- Gradient: $\nabla f_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$
- Search direction:
 $S_1 = -\nabla f_1 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$
- Line search:
 $f(\lambda) = f\left(\begin{bmatrix} -\lambda \\ \lambda \end{bmatrix}\right) = \lambda^2 - 2\lambda$
- Optimal step:
 $\frac{df}{d\lambda} = 2\lambda - 2 = 0 \Rightarrow \lambda_1^* = 1$
- Update: $X_2 = X_1 + \lambda_1^* S_1$
 $= \begin{bmatrix} 0 \\ 0 \end{bmatrix} + (1) \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$
- Function value: $f(X_2) = -1$

Method 2: Optimal Step Size (Line Search)

Minimize: $f(x_1, x_2) = x_1 - x_2 + 2x_1^2 + 2x_1x_2 + x_2^2$

Gradient: $\nabla f(x_1, x_2) = \begin{bmatrix} 1 + 4x_1 + 2x_2 \\ -1 + 2x_1 + 2x_2 \end{bmatrix} \Rightarrow \nabla f_2 = \begin{bmatrix} -1 \\ -1 \end{bmatrix}$

Iteration 2

- Current point: $X_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$
- Gradient: $\nabla f_2 = \begin{bmatrix} -1 \\ -1 \end{bmatrix}$
- Search direction:
 $S_2 = -\nabla f_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$
- Line search: $f(\lambda) = f\left(\begin{bmatrix} -1 + \lambda \\ 1 + \lambda \end{bmatrix}\right) = 5\lambda^2 - 2\lambda - 1$
- Optimal step:
 $\frac{df}{d\lambda} = 10\lambda - 2 = 0 \Rightarrow \lambda_2^* = 0.2$
- Update: $X_3 = X_2 + \lambda_2^* S_2$
 $= \begin{bmatrix} -1 \\ 1 \end{bmatrix} + (0.2) \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -0.8 \\ 1.2 \end{bmatrix}$
- Function value: $f(X_3) = -1.2$

Method 2: Optimal Step Size (Line Search)

Iteration 3

- Current point: $X_3 = \begin{bmatrix} -0.8 \\ 1.2 \end{bmatrix}$
- Gradient: $\nabla f_3 = \begin{bmatrix} 0.2 \\ -0.2 \end{bmatrix}$
- Search direction: $S_3 = -\nabla f_3 = \begin{bmatrix} -0.2 \\ 0.2 \end{bmatrix}$
- Line search: $f(\lambda) = f\left(\begin{bmatrix} -0.8 - 0.2\lambda \\ 1.2 + 0.2\lambda \end{bmatrix}\right) = 0.2\lambda^2 - 0.4\lambda - 1.2$
- Optimal step: $\frac{df}{d\lambda} = 0.4\lambda - 0.4 = 0 \Rightarrow \lambda_3^* = 1$
- Update: $X_4 = X_3 + \lambda_3^* S_3 = \begin{bmatrix} -1.0 \\ 1.4 \end{bmatrix}$
- Function value: $f(X_4) = -1.4$

Optimal Step Size Gradient Descent (Line Search) - Summary

Minimize: $f(x_1, x_2) = x_1 - x_2 + 2x_1^2 + 2x_1x_2 + x_2^2$

Gradient: $\nabla f(x_1, x_2) = \begin{bmatrix} 1 + 4x_1 + 2x_2 \\ -1 + 2x_1 + 2x_2 \end{bmatrix}$

i	X_i	∇f_i	S_i	λ_i^*	X_{i+1}
1	$\begin{bmatrix} 0 \\ 0 \end{bmatrix}$	$\begin{bmatrix} 1 \\ -1 \end{bmatrix}$	$\begin{bmatrix} -1 \\ 1 \end{bmatrix}$	1.0	$\begin{bmatrix} -1 \\ 1 \end{bmatrix}$
2	$\begin{bmatrix} -1 \\ 1 \end{bmatrix}$	$\begin{bmatrix} -1 \\ -1 \end{bmatrix}$	$\begin{bmatrix} 1 \\ 1 \end{bmatrix}$	0.2	$\begin{bmatrix} -0.8 \\ 1.2 \end{bmatrix}$
3	$\begin{bmatrix} -0.8 \\ 1.2 \end{bmatrix}$	$\begin{bmatrix} 0.2 \\ -0.2 \end{bmatrix}$	$\begin{bmatrix} -0.2 \\ 0.2 \end{bmatrix}$	1.0	$\begin{bmatrix} -1.0 \\ 1.4 \end{bmatrix}$

i	Line Search Function $f(\lambda)$	Optimality Condition	Function Value $f(X_{i+1})$
1	$\lambda^2 - 2\lambda$	$2\lambda - 2 = 0$	-1.0
2	$5\lambda^2 - 2\lambda - 1$	$10\lambda - 2 = 0$	-1.2
3	$0.2\lambda^2 - 0.4\lambda - 1.2$	$0.4\lambda - 0.4 = 0$	-1.4

Observations

- Variable step sizes: $\lambda^* = 1.0, 0.2, 1.0$ (adapts to local curvature)
- Much faster convergence: $-1.0 \rightarrow -1.2 \rightarrow -1.4$ vs fixed rate
- Line search finds optimal step along gradient direction each iteration
- More computational cost per iteration but fewer iterations needed

Comparative Analysis: Performance Comparison

Metric	Fixed Learning Rate	Optimal Step Size
After 3 iterations	$f = -0.3824$	$f = -1.2$
After 5 iterations	$f = -0.4563$	$f = -1.4$
Convergence speed	Slow	Rapid improvement
Step size	Constant: 0.1	Variable: 1.0, 0.2, 1.0
Computational cost	Low	High

Performance comparison after 3-5 iterations

Comparative Analysis: Trajectory Comparison

Fixed Learning Rate Path

$$X_1 = [0.0000, 0.0000], f = 0.0000$$

$$X_2 = [-0.1000, 0.1000], f = -0.1900$$

$$X_3 = [-0.1600, 0.1800], f = -0.3076$$

$$X_4 = [-0.1960, 0.2400], f = -0.3824$$

$$X_5 = [-0.2176, 0.2872], f = -0.4275$$

$$X_6 = [-0.2306, 0.3251], f = -0.4563$$

Optimal Step Size Path

$$X_1 = [0.0000, 0.0000], f = 0.0000$$

$$X_2 = [-1.0000, 1.0000], f = -1.0000$$

$$X_3 = [-0.8000, 1.2000], f = -1.2000$$

$$X_4 = [-1.0000, 1.4000], f = -1.4000$$

Comparative Analysis: Key Observations

- ① **Convergence Speed:** Optimal step size method converges much faster in terms of function value reduction.
- ② **Step Size Behavior:**
 - Fixed method uses conservative, constant steps
 - Optimal method adapts step sizes (sometimes taking large steps like $\lambda = 1$)
- ③ **Computational Trade-off:**
 - Fixed α : Cheap per iteration but needs more iterations
 - Optimal λ : Expensive per iteration but needs fewer iterations
- ④ **Practical Implications:**
 - For simple functions where line search is cheap, optimal step size is superior
 - For complex functions in machine learning, fixed/adaptive learning rates are more practical
- ⑤ **Stability:** Both methods show stable convergence without oscillation in this convex quadratic problem.

When to Use Each Method

Use Fixed Learning Rate when:

- Function evaluations are expensive
- Good learning rate can be estimated
- Implementing simple, scalable algorithms
- Working with stochastic/mini-batch variants

Use Optimal Step Size when:

- Function is cheap to evaluate multiple times
- High precision is required
- Analyzing theoretical properties
- Solving small-scale optimization problems

Key Takeaways

- Gradient descent is a fundamental optimization algorithm
- Fixed learning rate is simple but requires careful tuning
- Optimal step size converges faster but is computationally expensive
- Choice depends on problem characteristics and computational constraints
- Both methods follow the same fundamental principle: move against the gradient

General Update Rule

$$\mathbf{x}_{k+1} = \mathbf{x}_k - \lambda_k \nabla f(\mathbf{x}_k)$$

where λ_k can be fixed (α) or optimal (λ_k^*)